

Subject Description Form

Subject Code	EEE510
Subject Title	Introduction to Quantum Physics
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To provide students with a comprehensive understanding of the fundamental principles and mathematical formalism of quantum mechanics. And helps students know the application of quantum mechanics such as the exact solution for the spectrum of hydrogen atoms and perturbation theory for solving the quantum transition of the atom.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand exactly the fundamental concepts of quantum mechanics and the development history of quantum mechanics. - Know the essence of the Schrodinger equation and the uncertainty principle. - Learn how to solve Schrodinger equation for oscillator and hydrogen atom and other solvable situations. - Dip into the quantum angular momentum coupling theory for orbital and spin angular momentum of an atom. - Know the perturbation theory for explaining the time-dependent Schrodinger equation for stimulated transition.
Subject Synopsis/ Indicative Syllabus	Fundamental concept: Wave-particle duality and the uncertainty principle, Postulates of quantum mechanics, Dirac notation and bra-ket notation. Quantum dynamics: Time evolution of quantum systems, Schrödinger equation and its solutions, stationary states, and energy quantization. Quantum mechanical systems in three dimensions: Central potentials and the hydrogen atom, Harmonic oscillator, Measurement and probability in quantum mechanics. Angular momentum coupling theory: Angular momentum and spin, Angular momentum and spin, Zeeman effect. Applications of quantum mechanics: Quantum mechanics in condensed matter physics, Quantum mechanics in atomic and molecular physics, Quantum mechanics in nuclear and particle physics. Advanced topics: Time-independent and time-dependent perturbation theory. Density matrix formalism, Quantum field theory and relativistic quantum mechanics.

Teaching/Learning Methodology	Lecture: The Schrödinger equation and its application will be explained, accompanied by concrete examples illustrating the solution of the equation and the comparison between theory and experiment. Assignments will be provided to assess students' learning progress. Tutorial: A collection of problem sets and discussion topics will be provided. Students are strongly encouraged to attempt solving the problems prior to receiving the solutions.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
				a	b	c	d	e
	1. Continuous Assessments (Assignments and tests)		40%	√	√	√	√	√
	2. Examination		60%	√	√	√	√	√
	Total		100 %					
	Continuous assessments consist of assignments and mid-term test. These will assess the students’ understanding of basic concepts and principles in quantum mechanics. The examination will be conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.							
Student Study Effort Expected	Class contact:							
	▪ Lecture						33 Hrs	
	▪ Tutorial						6 Hrs	
	Other student study efforts:							
	▪ Reading and self-study						81 Hrs	
	The total student study effort						120 Hrs	
Reading List and References	Textbook - Shankar, Ramamurti. Principles of quantum mechanics. Springer Science & Business Media, 2012. References - Feynman, Richard P., Albert R. Hibbs, and Daniel F. Styer. Quantum mechanics and path integrals. Courier Corporation, 2010. - Zettili, Nouredine. Quantum mechanics: concepts and applications. 2009. - Townsend, John S. A modern approach to quantum mechanics. University Science Books, 2000.							